

GET it digital

Modul 2: Energy and power



Stand: 18. Februar 2026



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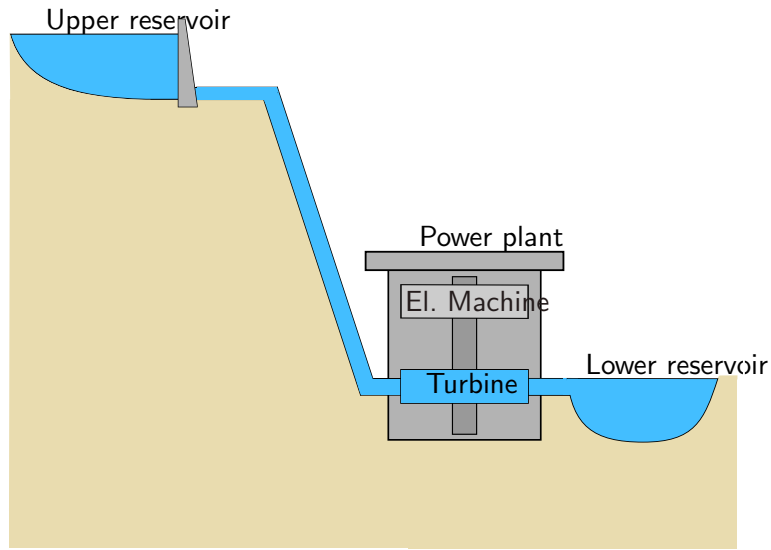
<https://getitdigital.uni-wuppertal.de/module/modul-2-energie-und-leistung>

Learning objectives: Energy and electrical power

The students

- ▶ can name energy, electrical work, power and efficiency.
- ▶ understand how energy is converted and used in electrical systems.
- ▶ can calculate electrical work, power and efficiency.
- ▶ can analyse the efficiency of energy conversion systems.

Pumpspeicherkraftwerk



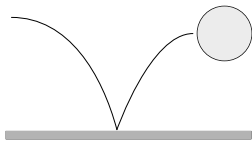
Key point:



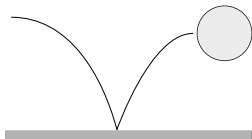
Key point:

Energy can be stored, work is done.

The law of conservation of energy



The law of conservation of energy



Key point:

Energy is a conserved quantity.
Energy can neither be created nor destroyed.
Energy can only change its form.

Learning objectives: The Electric Charge

The students can

- ▶ describe the properties of electric charges and related physical phenomena
- ▶ describe electric fields and calculate them for simple charge arrangements
- ▶ Calculate forces on charges using Coulomb's law

Key point: The forms of energy

Kinetic energy (energy of motion)

Potential energy (energy of position)

Thermal energy (heat energy)

Electrical energy

Chemical energy

Nuclear energy

Radiant energy

- Is generally defined as the integrated distance between two points P_1 and P_2 via a force $\vec{F}$

$$E = \int_{P_1}^{P_2} \vec{F} \cdot d\vec{s} \quad [E] = 1 \text{ Joule} = 1 \text{ J} = 1 \text{ Nm}$$

- ▶ An electric force $\vec{F}_{\text{el}}$ is present when a charge Q is exposed to an electric field $\vec{E}$.

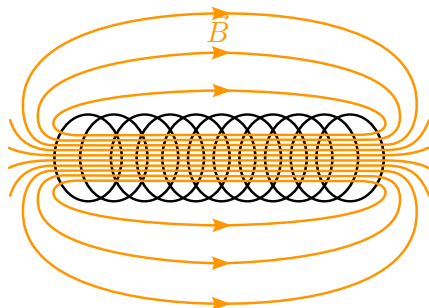
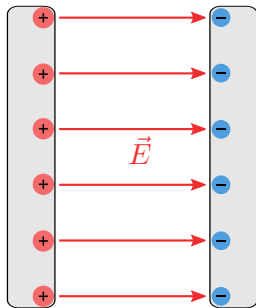
$$\vec{F}_{\text{el}} = Q \cdot \vec{E}$$

- ▶ ...

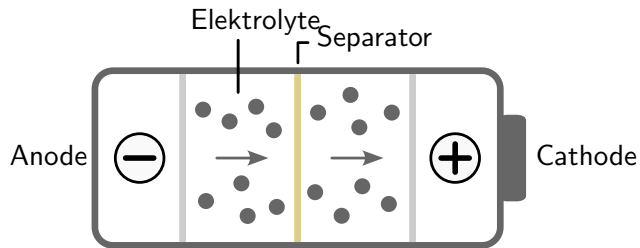
$$E_{\text{el}} = Q \cdot \int_{P_1}^{P_2} \vec{E} \cdot d\vec{s}$$

$$E_{\text{el}} = Q \cdot U \quad [E] = 1 \text{ Joule} = 1 \text{ J} = 1 \text{ Ws}$$

Electrical energy storage devices



Chemical energy storage



- ▶ Work is the change of energy form.

$$W = \Delta E \quad [W] = 1 \text{ Joule} = 1 \text{ J} = 1 \text{ Nm}$$

- ▶ ...

$$W = - \int_{P_1}^{P_2} \vec{F} \cdot d\vec{s}$$

Key point:

The sign of the work is context-dependent: Work is positive when energy is added to a system and negative when energy is removed from a system.

$$\vec{E} = \frac{\vec{F}_{\text{el}}}{Q} \quad \Rightarrow \quad \vec{F}_{\text{el}} = Q \cdot \vec{E}$$

$$W_{\text{el}} = \int_{P_1}^{P_2} -Q \cdot \vec{E} \cdot d\vec{s} \quad \Rightarrow \quad W_{\text{el}} = -Q \int_{P_1}^{P_2} \vec{E} \cdot d\vec{s}$$

$$W_{\text{el}} = Q \int_{P_1}^{P_2} \vec{E} \cdot d\vec{s} \quad \Rightarrow \quad W_{\text{el}} = Q \cdot U$$

$$I = \frac{dQ}{dt} \quad \Rightarrow \quad Q = I \cdot t$$

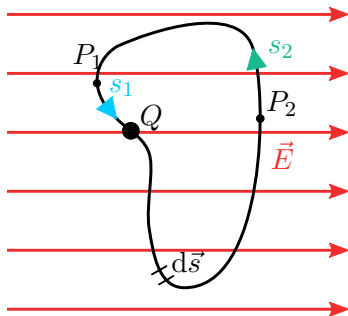
When inserted, this now results in the following expression

$$W_{\text{el}} = U \cdot I \cdot t \quad [W_{\text{el}}] = 1 \text{ Joule} = 1 \text{ J} = 1 \text{ Ws}$$

Key point:

- ▶ A certain amount of time is always required to complete a task.
- ▶ A momentary state is always the actual state of energy distribution.
- ▶ Performing electrical work always requires a flow of current.

Path integral of electrical work



$$|W_{\text{el}}| = \left| -Q \cdot \int_{s_1} \vec{E} \cdot d\vec{s} \right| = \left| -Q \cdot \int_{s_2} \vec{E} \cdot d\vec{s} \right|$$

$$W_{\text{el}} = -Q \cdot \int_{s_1} \vec{E} \cdot d\vec{s} - Q \cdot \int_{s_2} \vec{E} \cdot d\vec{s} = -Q \cdot \oint \vec{E} \cdot d\vec{s} = 0 \rightarrow \oint \vec{E} \cdot d\vec{s} = 0$$

Key point:

The line integral of a charge in a static electric field is always zero, regardless of the amount of charge.

$$W = \int P \cdot dt$$

By rearranging the formula, we see that power P is the change in work over time.

$$P = \frac{dW}{dt}$$

If the work remains constant over time, the following expression results:

$$P = \frac{W_{\text{el}}}{t} = \frac{U \cdot I \cdot \cancel{t}}{\cancel{t}} = U \cdot I \quad [P] = 1 \text{ Watt} = 1 \text{ W}$$

Types	Potential value	Flow rate	Formula
Electric	Voltage (U)	Current (I)	$P_{\text{el}} = U \cdot I$
Translational	Force (F)	Velocity (v)	$P_{\text{tr}} = F \cdot v$
Rotational	Moment (M)	Angular velocity (ω)	$P_{\text{rot}} = M \cdot \omega$
Thermal	Temperature diff. (ΔT)	Heat transfer ($k \cdot A$)	$P_{\text{th}} = \Delta T \cdot k \cdot A$
Fluidic	Pressure (p)	Volume flow ($\dot{V}$)	$P_{\text{fl}} = p \cdot \dot{V}$

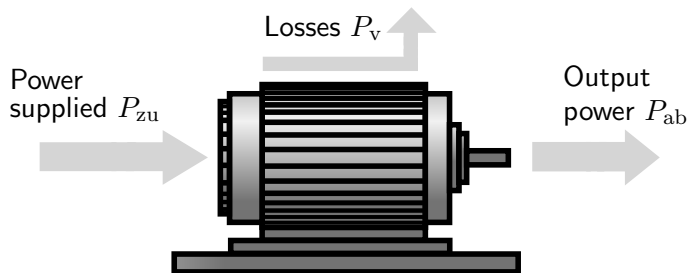


Key point:

Electrical work measures the energy transferred, while electrical power indicates the transfer speed.

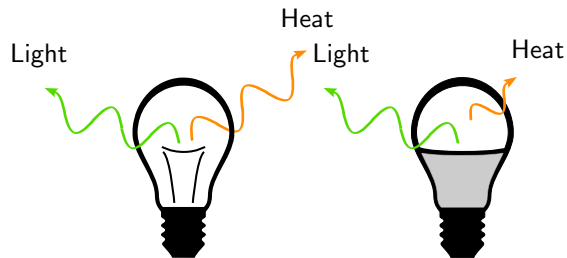
The efficiency

Efficiency describes the ratio of output power to input power.

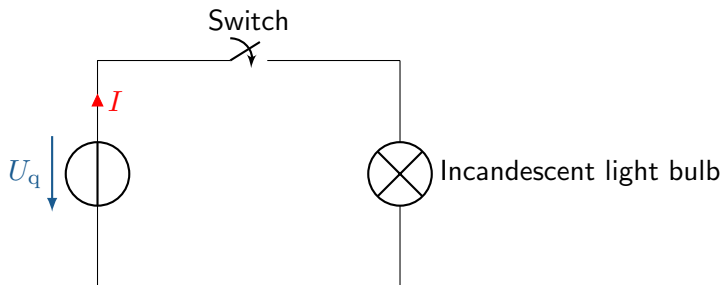


$$\eta = \frac{P_{out}}{P_{in}} \cdot 100\%$$

The efficiency



The efficiency



The efficiency

